Biomechanics to Basketball Intelligence: Constrained Counterfactual Coaching with 3D Visualization

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PEAK PERFORMANCE PROJECT

UC SANTA BARBARA | Data Science Initiative

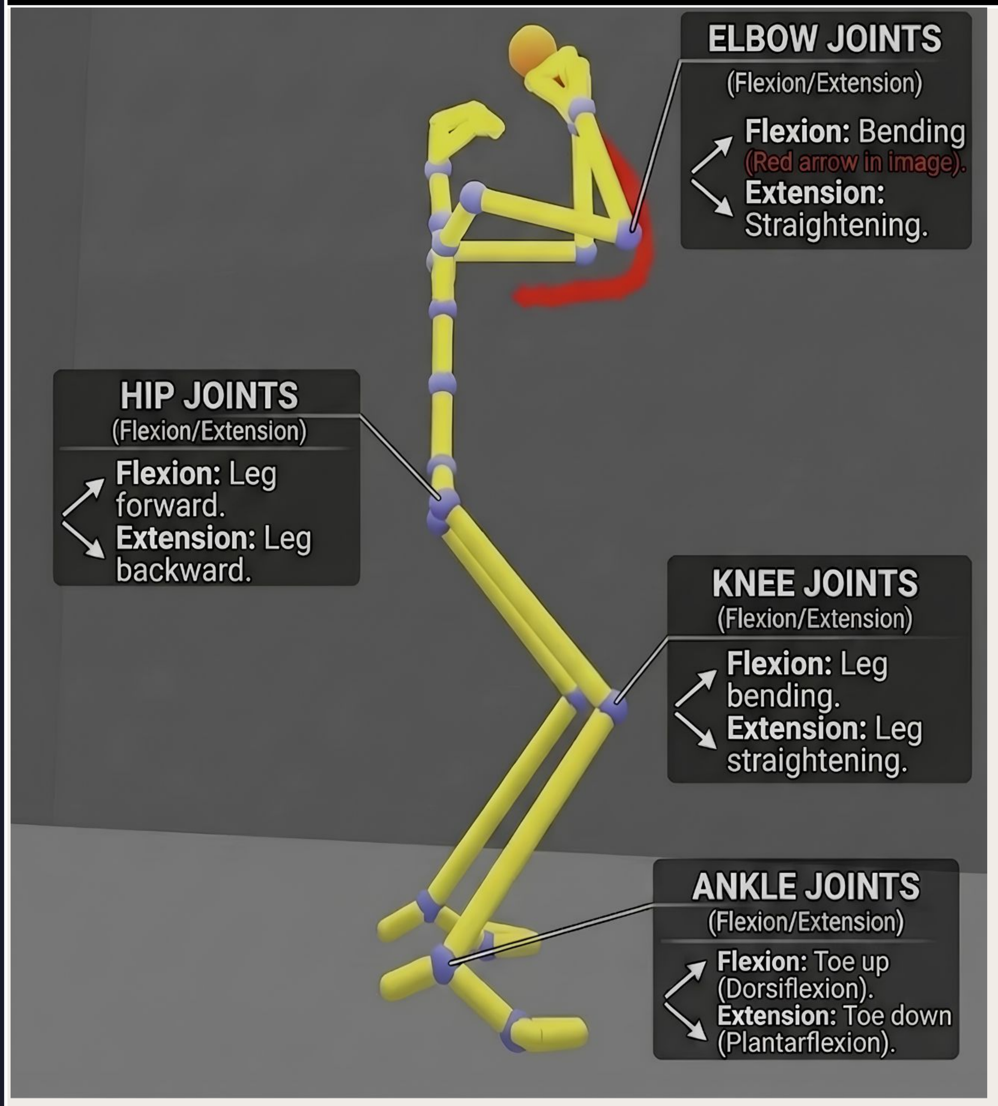
Background

Traditional basketball coaching relies on make/miss percentages and film study, metrics that show *what* is happening but not *why*. Two players shooting from three may need entirely different mechanical corrections, but outcome data alone can't distinguish between them.

P3 Peak Performance Project has collected shooting data using state of the art biomechanical tools. The dataset contains thousands of shots from hundreds of players, with more than 200 variables per shot measuring joint angles, velocities, alignment, and timing across multiple phases of the shooting motion.

Our goal: build an interactive tool that translates machine learning insights from this data into visual, player-specific biomechanical feedback.

Dataset



Thousands of shots · 100+ players · 200+ variables

Each shot is segmented into four biomechanical phases, Loading, Upward transition, Acceleration, and Release, with variables measured at each phase boundary.

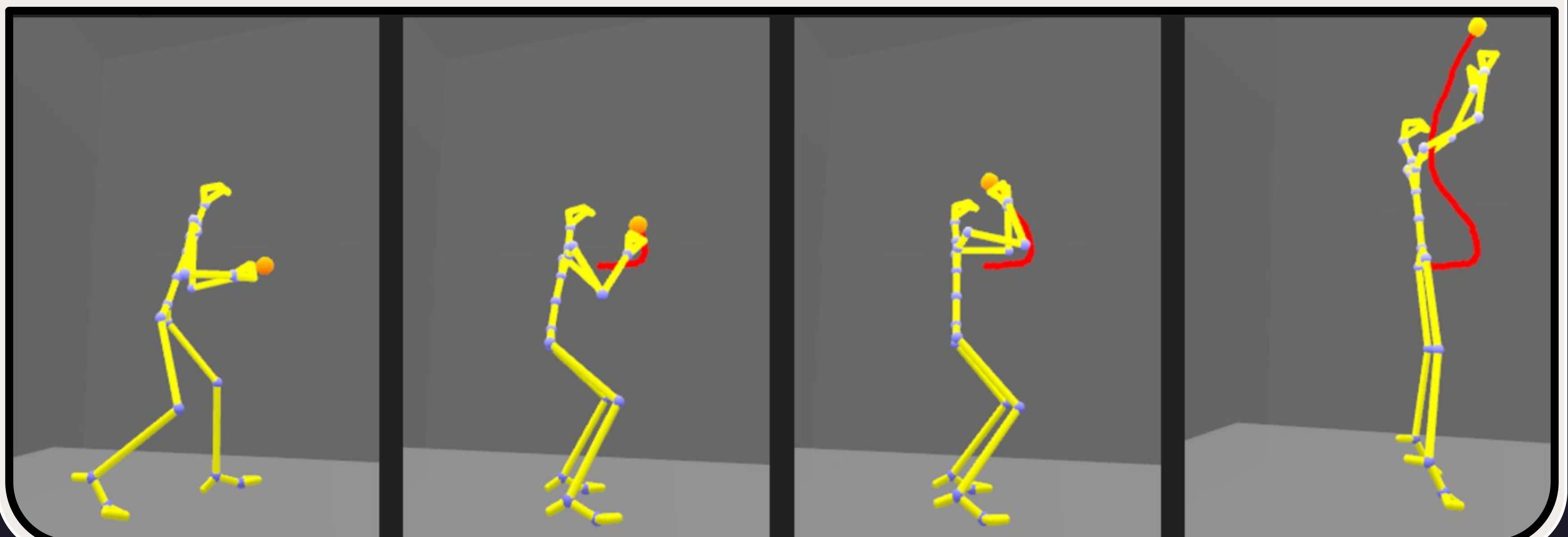
Joint angles: knee, hip, shoulder, elbow, and ankle flexion/extension for dominant and non-dominant sides

Ball flight: release velocity, launch angle, entry angle, max arc height, rim depth and left/right entry

Body dynamics: torso velocity, torso alignment, stance width, center of mass displacement

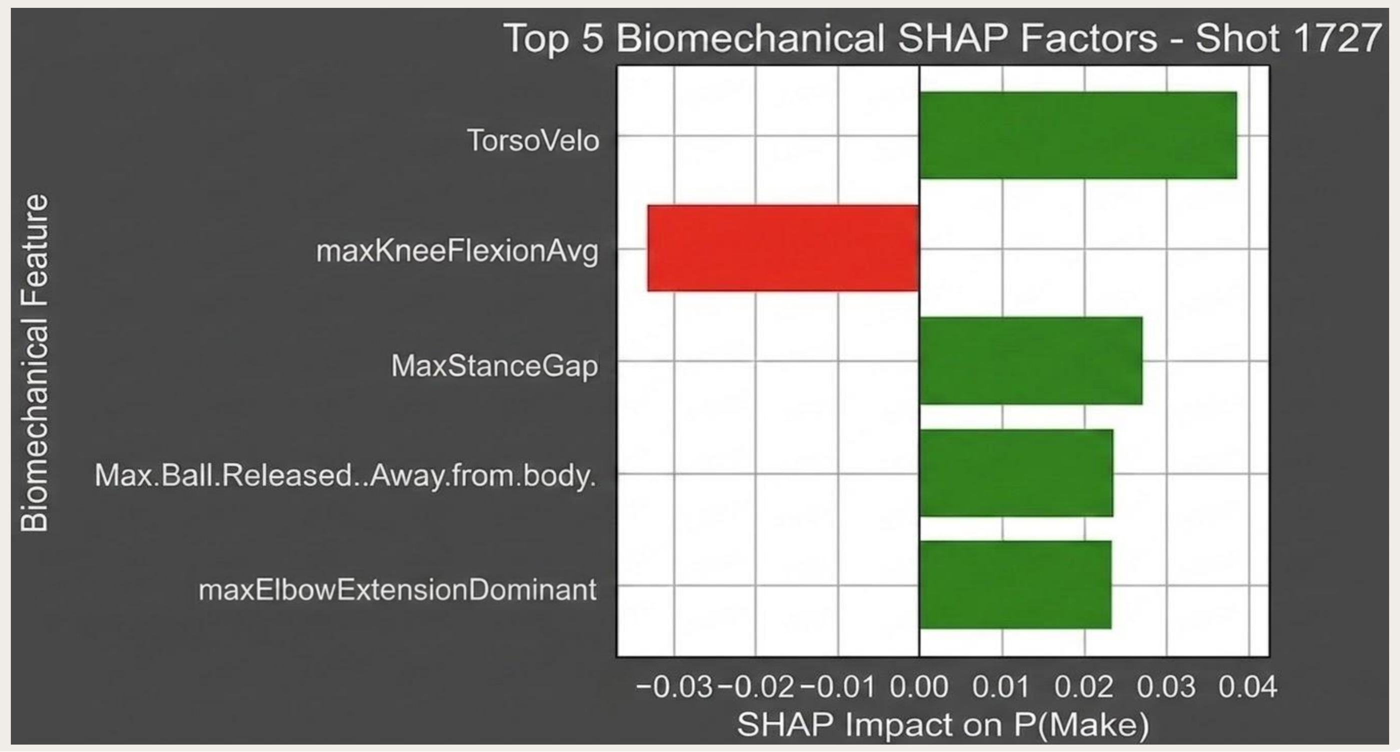
Timing: phase durations, time to peak flexion/extension events

Data preparation:
Left-handed players reflected to right-handed frame ("Dominant" = shooting side)
Release height scaled by player height; capped at 500 shots per player
Per-player z-scores computed to capture individual consistency patterns

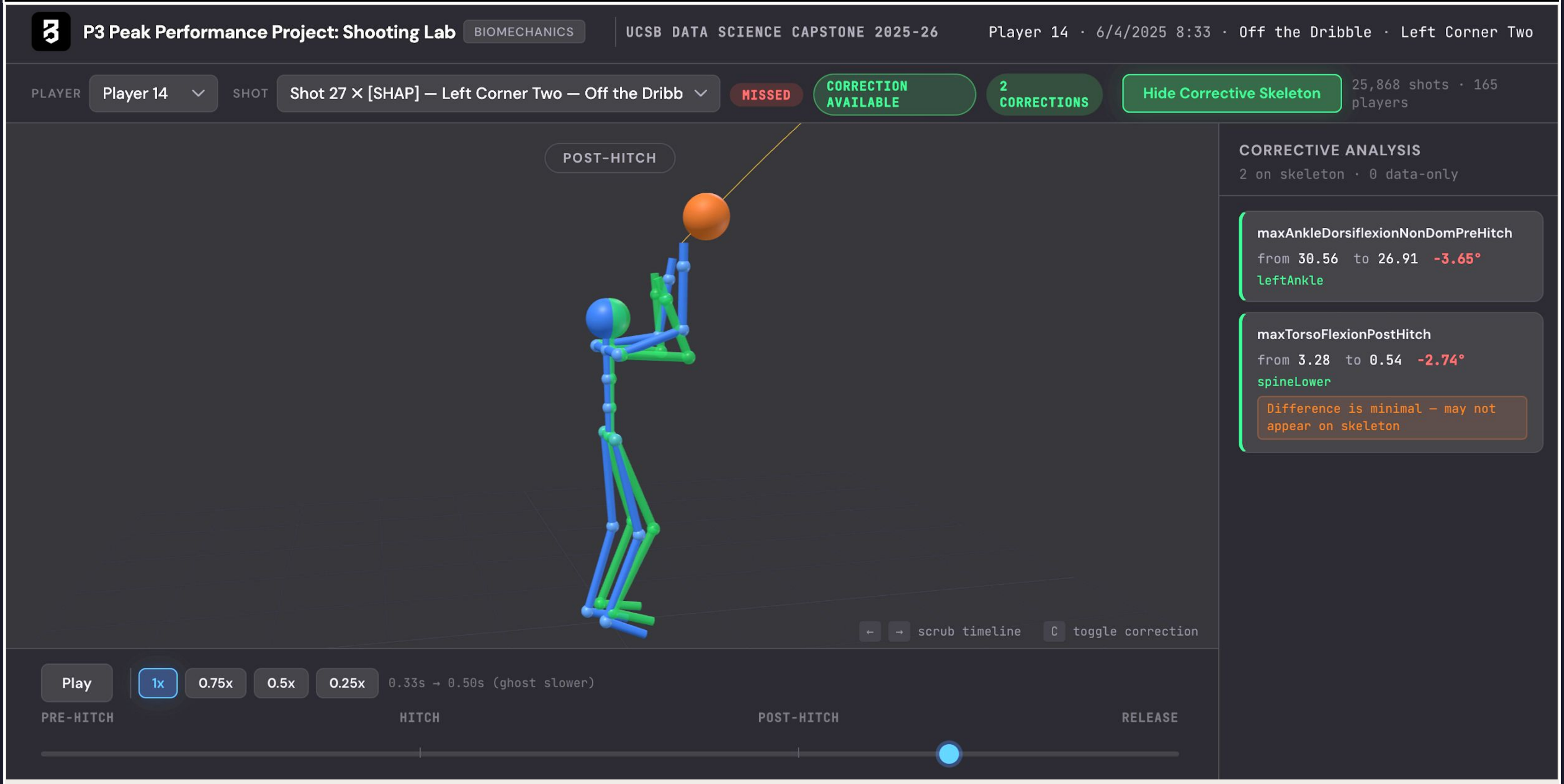


Methodology

- **Stage 1: LightGBM Modeling** Multi-target LightGBM models predict ball flight metrics from biomechanical inputs (R^2 = 0.53–0.88), capturing how mechanics influence shot quality. A separate binary classifier predicts make/miss and drives all downstream counterfactual recommendations.
- **Stage 2: SHAP Explainability** Per-shot SHAP values identify the 3–5 biomechanical features most hurting each missed shot. Features with negative SHAP impact (pulling the prediction toward "miss") are ranked by magnitude and filtered to coachable variables only.
- **Stage 3: Counterfactual Corrections** DiCE Random search then finds the smallest pose change that flips the classifier's prediction, with each feature's range bounded by anatomical limits, per-player feasibility, and coaching-scale caps enforced upstream during sampling.



Results



Interactive 3D biomechanics visualizer showing a player's missed shot (blue skeleton) with DiCE counterfactual corrections overlaid as a ghost skeleton (green). The sidebar displays the top biomechanical corrections with original and recommended values. Animation playback speed reflects velocity correction deltas from the model.

Built with: Three.js · LightGBM · SHAP · PapaParse · Vite · Python / pandas

Key Findings

- SHAP corrections are highly player-specific, two players with identical shooting percentages receive different top-5 corrective cues, confirming that outcome metrics alone are insufficient for mechanical diagnosis.
- Corrections are biomechanically targeted: feature windows are bounded upstream by anatomical limits and per-shot delta caps, so DiCE only recommends subtle, realistic changes.
- The pipeline identifies shot-type-specific corrections; a player's off-the-dribble misses receive different recommendations than their catch-and-shoot misses, reflecting the distinct mechanics of each shot type.

Future Work

- Accept raw biomechanical input directly for real-time shot analysis without pre-export
- Display real ball-flight trajectory with visualization of ball going in or out of the hoop
- Extend to other movement patterns beyond shooting (e.g. athleticism, injury risk)
- Deploy within P3's production platform

References and Acknowledgements

[1] Ramaravind, K., Mothilal, R. K., & Sharma, A. (2020). Source code for dice_ml.explainer_interfaces.dice_random. *DiCE 0.12 documentation.*

[2] Lundberg, S. M. (2025). SHAP: SHapley Additive exPlanations. *SHAP documentation.* https://shap.readthedocs.io/en/latest/

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